

What Predicts NFL Draft Slot?

Statistically significant pre-draft predictors, from 8,660 NFL Combine invitees (2000–2026)

DraftDay research · September 2026

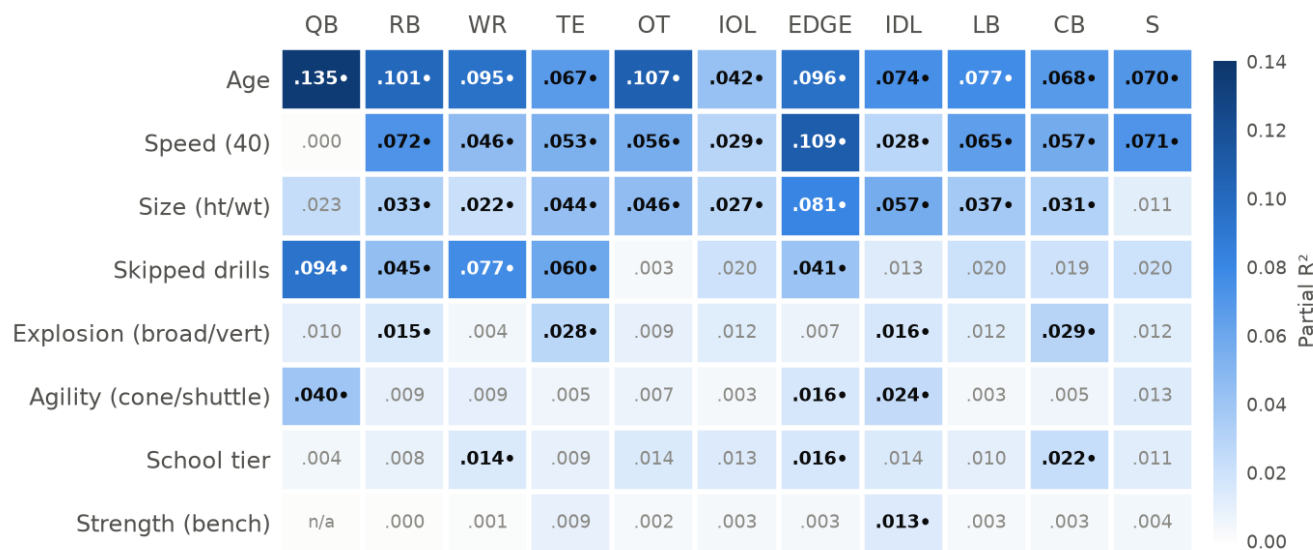
The short answer

Four things predict where a prospect is drafted, consistently and in roughly this order:

- **Age at draft.** Younger players go earlier, at every one of the 11 positions.
- **Speed for size.** The 40, best expressed as weight-adjusted speed score. The top athletic trait at 10 of 11 positions.
- **Size for the position.** Matters most at EDGE, interior D-line, offensive tackle and TE.
- **College production,** for QB, TE, RB and WR only. It adds significant signal on top of everything else.

Vertical jump, short shuttle and bench press add almost nothing to *where* a player goes once the rest is known. Public measurables plus production predict held-out draft years with a rank correlation of only about 0.3–0.5, so most of draft slot is driven by scouting information, which DraftDay captures through the mock-draft consensus. Use these findings as priors and supporting features, not as a replacement for consensus.

Trait importance at a glance



How much each trait group explains about draft slot among drafted players, beyond all other traits (partial R² from drop-one-group F-tests, 2000–2026). **Bold with •** = significant after false-discovery-rate correction ($q < 0.05$); gray = not significant. QBs are excluded from the bench model because they almost never bench.

The question and the data

College prospects have thin data. Players who have already been through the draft have a lot. This study asks which traits, measured *before* the draft, are statistically significant predictors of where a player gets picked, so the answers can drive DraftDay's prospect and player models.

The population is every NFL Combine invitee from 2000 to 2026, drafted or not (65% were drafted), with kickers, punters and long snappers removed. Keeping undrafted invitees separates two questions: *whether* a prospect is drafted, and *where*.

Source	What it provides	Coverage
nflverse combine	Height, weight, 40, bench, vertical, broad, 3-cone, shuttle, school, draft pick	All invitees 2000–2026
nflverse draft picks / players	Pick, age, birth date	Players who reached the NFL
cfbfastR team info	Conference and FBS/FCS status by season	99.5% of invitees matched
cfbfastR play-level stats	College passing, rushing, receiving and defensive production	2015+ draft classes; 92–98% of skill players and DBs, ~78% of DL, no OL

How significance was tested

Each position is analyzed separately. Drill results are standardized within position and oriented so positive always means better. A trait only counts as a predictor if four independent tests broadly agree:

- **Rank correlation** (Spearman) with draft position.
- **Multivariate regression** with robust (HC3) standard errors, controlling for every other trait and draft year. Three outcomes: slot among drafted players (–log pick), whether drafted (logit), and draft capital with undrafted set to pick 300.
- **Nested F-tests** that drop one trait group at a time, measuring what each group adds (partial R^2).
- **Out-of-sample checks:** gradient boosting trained on all other draft years predicts each held-out year; shuffling a trait group and measuring the accuracy lost gives its importance.

All p-values are corrected for testing many traits at once (Benjamini-Hochberg false discovery rate). **Leakage guard:** birth dates exist almost only for players who reached the NFL, so a missing age nearly means “undrafted.” Age is therefore used only in the drafted-players model.

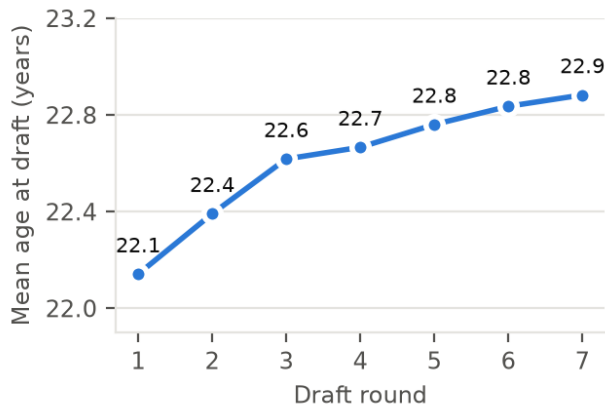
Findings

1. Age is the most consistent predictor of slot

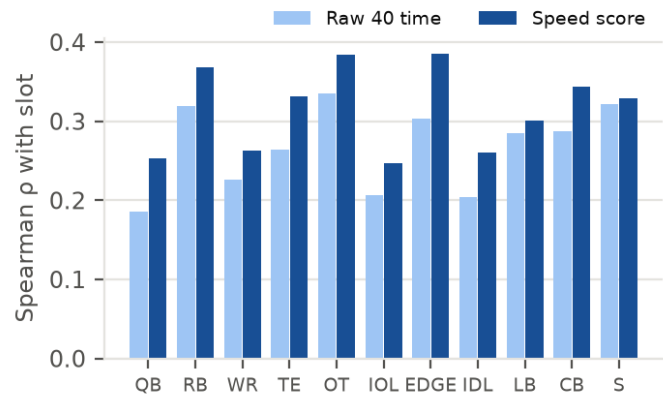
Age is significant at all 11 positions ($q < 0.001$), explaining 4% (IOL) to 13.5% (QB) of slot variation on its own margin, and it is the largest single trait group at 9 of 11 positions. One standard deviation of age, about 0.9 years, moves a pick about 30% later. Part of this is reverse causality (top prospects declare early), but it is a clean pre-draft signal the current prospect model does not use.

2. Speed, adjusted for size, is the dominant athletic trait

The 40 is significant for slot at every position except QB. In the out-of-sample test, shuffling speed costs 0.11–0.31 of rank correlation, the biggest loss of any group at 10 of 11 positions. Weight alone barely correlates with slot, but becomes strongly positive once the 40 is in the model: teams pay for speed at a *given size*. Speed score ($\text{weight} \times 200 / 40\text{-time}^4$) captures this and beats the raw 40 at all 11 positions.



Mean age at draft falls steadily from round 7 to round 1 (drafted players, 2000–2026).



Rank correlation with draft slot: speed score (dark) vs raw 40 time (light), by position.

3. Size matters most in the trenches and at EDGE

Height and weight together are significant for slot at 9 of 11 positions, largest at EDGE (8.1%), interior D-line (5.7%), offensive tackle (4.6%) and TE (4.4%). QB height matters for *getting drafted* but not for where.

4. The secondary drills are weak

- **Broad jump** is significant league-wide and at CB and TE. The **vertical** is never significant at any position once the broad and 40 are known.
- **3-cone** matters for QB, EDGE and interior D-line, and out of sample for the offensive line.
- **Short shuttle** predicts being drafted but not where.
- **Bench press** has no significant effect on slot at any position except interior D-line (1.3%).

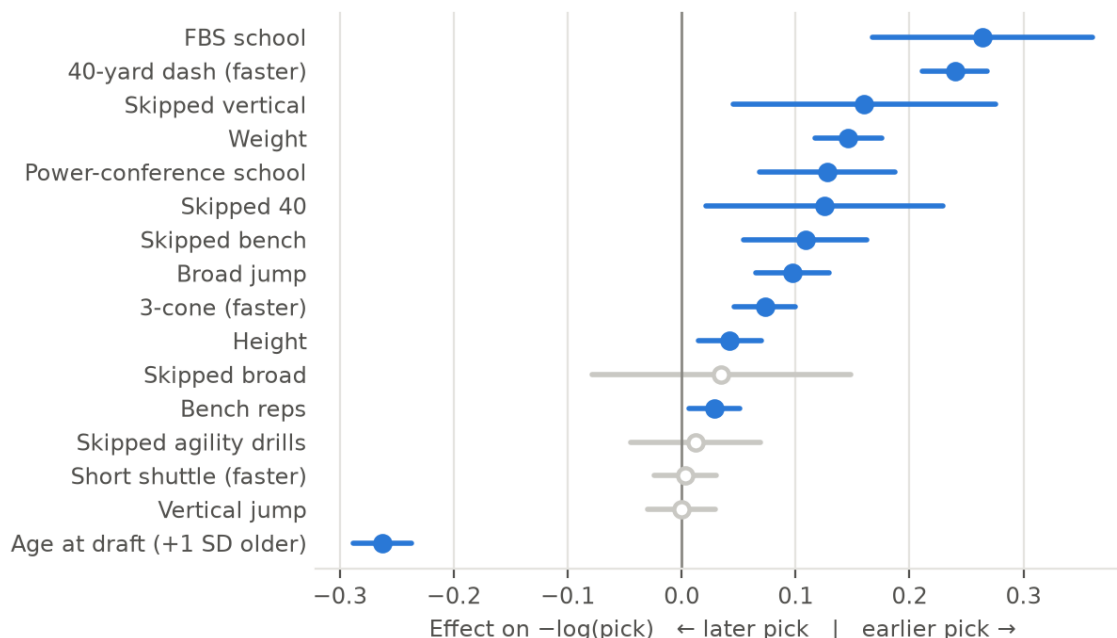
5. School tier matters more for being drafted than for where

FBS and Power-conference players are much more likely to be drafted, but among drafted players school tier explains at most 2.2% of slot.

6. Skipping drills signals top-prospect status

At QB, WR, TE, RB and EDGE, skipping drills is associated with *earlier* picks, likely because consensus top prospects sit out. The combine file shows almost no missing 40 times before 2017, so this signal comes mainly from 2017–2026. Treat it as a proxy for scouting consensus, not a trait.

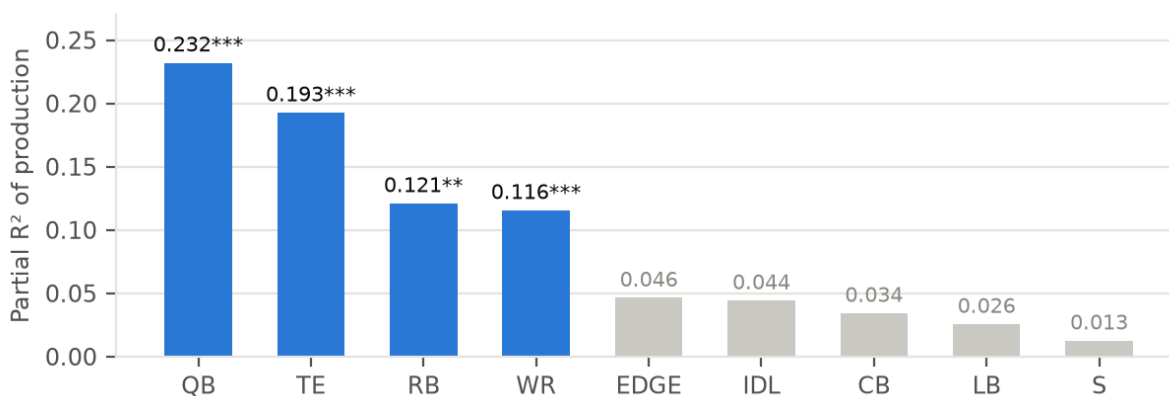
League-wide effects



All positions pooled with position fixed effects, drafted players. Dots show the effect on $-\log(\text{pick})$ with 95% confidence intervals; filled blue = significant ($q < 0.05$), hollow gray = not. Measurements are per 1 SD within position; school and “skipped” flags are 0/1, so their bars are on a different scale.

7. College production adds signal for QB, TE, RB and WR, not for defenders

Production was tested on top of combine, age and school, using 2015+ classes with linked college stats. Rates and team shares are used instead of raw totals because the play-level feed covers only about 70-85% of games.



Share of slot variation explained by production beyond all other traits, 2015+ drafted players. Blue = significant (* $q < 0.05$, ** $q < 0.01$, *** $q < 0.001$).

Pos	Metrics that carry the signal (rank correlation with slot)	Held-out accuracy gain
QB	Final-year yards/attempt +0.32, TD rate +0.24, fewer college seasons	+0.16 (not significant, n = 121)
TE	Receiving yards/game +0.34, share of team receiving yards +0.28, yards/target +0.23	+0.15 **
RB	TDs/game +0.36, rushing yards/game +0.29, yards/carry +0.25	+0.27 **
WR	Dominator rating +0.25, share of team receiving yards +0.25, yards/target +0.19, younger breakout age +0.18	+0.21 ***
EDGE	Sacks/game +0.20 on its own, but absorbed by athletic traits	+0.12 (q = 0.052)
DL, LB, DB	Safety INTs/game +0.22 on its own only; CB ball production shows nothing	Not significant

8. The relationships are stable over time

Comparing 2000–2012 with 2013–2026, only 1 of 166 trait effects changed significantly (interior D-line 3-cone faded to zero). The findings can be applied to current draft classes.

9. The ceiling is modest

Predicting each draft year from the others, the traits reach a rank correlation of 0.32–0.43 with draft capital from combine and school alone, and up to 0.52 with production. Among drafted players it is 0.19–0.49. Film, medicals and interviews explain the rest.

Position	Combine + school (2000+)	Drafted only	With production (2015+)	Drafted only
QB	0.37	0.30	0.46	0.36
RB	0.43	0.35	0.52	0.34
WR	0.43	0.31	0.47	0.49
TE	0.36	0.25	0.39	0.36
OT	0.43	0.35	–	–
IOL	0.36	0.22	–	–
EDGE	0.43	0.38	0.42	0.35
IDL	0.32	0.24	0.36	0.22
LB	0.40	0.30	0.44	0.35
CB	0.42	0.29	0.31	0.19
S	0.39	0.32	0.38	0.33

Rank correlation between held-out predictions and actual draft capital (undrafted = pick 300). The two column pairs use different samples, so compare them loosely.

Using this in DraftDay

Per-position models, with their exact preprocessing, are exported to `backend/artifacts/draft_slot/position_priors.json` and applied by `score_prospects()` in `backend/app/draft_slot/priors.py`.

Variant	Predicts	Needs	Use for
athletic	Draft capital	Combine + school	ProspectFeature.athletic_score, for any invitee
slot	Pick, among drafted	+ age	Expected pick; fallback consensus rank for prospects with thin mock coverage
production	Pick, among drafted	+ college stats (2015+)	ProspectFeature.production_score for QB, RB, WR, TE

Recommendations

- **Add age at draft** to the prospect and player features. It is the strongest signal the models currently lack.
- **Build athletic_score from position-specific weights:** speed score first, then size, broad jump and 3-cone. Drop the vertical and bench (except bench for interior D-line).
- **Build production_score only for QB, RB, WR and TE** using the metrics above; defensive production is not worth model capacity.
- **Blend, don't replace.** Combine the expected pick with the mock-draft consensus rank, giving the prior more weight when mock coverage is sparse (Day 3 prospects).

Limitations

- Combine invitees only; drafted players who were not invited (roughly 10–15% of picks) are missing.
- Before 2017 the source appears to fill missing 40 times with pro-day results; other drills are combine-only.
- No injury history, medical flags, recruiting rankings, PFF grades or historical big boards, so the study cannot yet measure what traits add *beyond* the market.
- Birth dates for undrafted invitees are missing, which limits age to the drafted-players model.
- No college production for offensive linemen.

Reproduce with `python -m backend.app.draft_slot.cli`. Full result tables are in `backend/artifacts/draft_slot/report.md`. Data from `nflverse` and `sportsdataverse cfbfastR-data`.